

10.7.51

EE25BTECH11002 - Achat Parth Kalpesh

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Question

Find the coordinates of all the points **P** on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, for which the area of the triangle PON is maximum, where **O** denotes the origin and **N**, the foot of the perpendicular from **O** to the tangent at **P**

Solution

The equation of the ellipse is given as

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (1)$$

$$\mathbf{V} = \begin{pmatrix} \frac{1}{a^2} & 0 \\ 0 & \frac{1}{b^2} \end{pmatrix}, \mathbf{u} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, f = -1 \quad (2)$$

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} = 1 \quad (3)$$

The equation of a tangent to $\mathbf{P}$ is

$$(\mathbf{V}\mathbf{P} + \mathbf{u})^\top \mathbf{x} + \mathbf{u}^\top \mathbf{P} + f = 0 \quad (4)$$

$$(\mathbf{V}\mathbf{P})^\top \mathbf{x} + f = 0 \quad (5)$$

As $\mathbf{N}$ is the foot of the perpendicular from $\mathbf{O}$ to the tangent at $\mathbf{P}$

$$\mathbf{N} = \mathbf{0} - \frac{(\mathbf{V}\mathbf{P})^\top \mathbf{0} + f}{\|\mathbf{V}\mathbf{P}\|} \mathbf{V}\mathbf{P} \quad (6)$$

$$\mathbf{N} = \frac{\mathbf{V}\mathbf{P}}{\|\mathbf{V}\mathbf{P}\|} \quad (7)$$

Solution

The cross product of two 2D vectors $\mathbf{x}$ and $\mathbf{y}$ can be written in matrix form using the skew-symmetric matrix

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (8)$$

Hence,

$$\mathbf{x} \times \mathbf{y} = \mathbf{x}^\top \mathbf{J} \mathbf{y}. \quad (9)$$

The area of the triangle $\triangle PON$ is given by

$$\text{Area}(PON) = \frac{1}{2} \frac{|\mathbf{P}^\top \mathbf{J} \mathbf{V} \mathbf{P}|}{\|\mathbf{V} \mathbf{P}\|^2}, \quad (10)$$

Solution

Let

$$\mathbf{Q} = \mathbf{V}\mathbf{P} \quad (11)$$

$$\mathbf{P} = \mathbf{V}^{-1}\mathbf{Q}. \quad (12)$$

Then, the area becomes

$$\text{Area}(PON) = \frac{1}{2} \frac{|(\mathbf{V}^{-1}\mathbf{Q})^\top \mathbf{J} \mathbf{Q}|}{\|\mathbf{Q}\|^2} \quad (13)$$

$$= \frac{1}{2} \frac{|\mathbf{Q}^\top (\mathbf{V}^{-1}\mathbf{J}) \mathbf{Q}|}{\|\mathbf{Q}\|^2}. \quad (14)$$

Since $\mathbf{Q}^\top \mathbf{M} \mathbf{Q}$ depends only on the symmetric part of $\mathbf{M}$, define

$$\mathbf{S} = \frac{\mathbf{V}^{-1}\mathbf{J} + (\mathbf{V}^{-1}\mathbf{J})^\top}{2} \quad (15)$$

Solution

Hence,

$$\text{Area}(\mathbf{Q}) = \frac{1}{2} \frac{|\mathbf{Q}^\top \mathbf{S} \mathbf{Q}|}{\|\mathbf{Q}\|^2}. \quad (16)$$

Now,

$$\mathbf{V}^{-1} = \begin{pmatrix} a^2 & 0 \\ 0 & b^2 \end{pmatrix}, \quad \mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (17)$$

Thus,

$$\mathbf{V}^{-1} \mathbf{J} = \begin{pmatrix} 0 & a^2 \\ -b^2 & 0 \end{pmatrix}, \quad (\mathbf{V}^{-1} \mathbf{J})^\top = \begin{pmatrix} 0 & -b^2 \\ a^2 & 0 \end{pmatrix}. \quad (18)$$

Solution

So,

$$\mathbf{S} = \frac{1}{2} \left(\mathbf{V}^{-1} \mathbf{J} + (\mathbf{V}^{-1} \mathbf{J})^{\top} \right) \quad (19)$$

$$= \frac{1}{2} \begin{pmatrix} 0 & a^2 - b^2 \\ a^2 - b^2 & 0 \end{pmatrix} \quad (20)$$

$$= \frac{a^2 - b^2}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (21)$$

The eigenvalues and eigenvectors of $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ are

$$\lambda_1 = 1, \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \lambda_2 = -1, \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (22)$$

Solution

Hence, the eigenvalues of $\mathbf{S}$ are $\pm \frac{a^2 - b^2}{2}$, and the maximum absolute eigenvalue is

$$\lambda_{\max} = \frac{|a^2 - b^2|}{2}. \quad (23)$$

The maximum area is therefore

$$\text{Area}_{\max} = \frac{|a^2 - b^2|}{4} \quad (24)$$

The corresponding eigenvector is

$$\mathbf{Q} \propto \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (25)$$

Hence,

$$\mathbf{P} = \mathbf{V}^{-1} \mathbf{Q} \propto \begin{pmatrix} a^2 \\ b^2 \end{pmatrix}. \quad (26)$$

Imposing the ellipse condition $\mathbf{P}^T \mathbf{V} \mathbf{P} = 1$,

Solution

Hence,

$$\mathbf{P} = \mathbf{V}^{-1}\mathbf{Q} \propto \begin{pmatrix} a^2 \\ b^2 \end{pmatrix}. \quad (30)$$

Imposing the ellipse condition $\mathbf{P}^\top \mathbf{V} \mathbf{P} = 1$,

$$1 = \lambda^2 (a^2 + b^2) \implies \lambda = \pm \frac{1}{\sqrt{a^2 + b^2}}. \quad (31)$$

Thus, the required points are

$$\mathbf{P} = \pm \frac{1}{\sqrt{a^2 + b^2}} \begin{pmatrix} a^2 \\ b^2 \end{pmatrix} \quad (32)$$

and the maximum area of $\triangle PON$ is

$$\text{Area}_{\max} = \frac{|a^2 - b^2|}{4} \quad (33)$$

```
#include <stdio.h>
#include <math.h>
#include <stdlib.h>

// ----- Define structure -----
typedef struct {
    double x;
    double y;
} Vec2;

// ----- Compute V @ P -----
void tangent_normal_at(double a, double b, Vec2 P, Vec2 *n) {
    n->x = P.x / (a * a);
    n->y = P.y / (b * b);
}
```

```
// ----- Foot of perpendicular from origin to line nx = 1
// -----
void foot_from_origin_to_line(Vec2 n, double c, Vec2 *N) {
    double denom = n.x * n.x + n.y * n.y;
    N->x = (c / denom) * n.x;
    N->y = (c / denom) * n.y;
}

// ----- Cross product area -----
double area_triangle(Vec2 P, Vec2 O, Vec2 N) {
    double cross = P.x * N.y - P.y * N.x;
    return 0.5 * fabs(cross);
}

// ----- Compute optimal P points -----
void compute_P_points(double a, double b, Vec2 *P_plus, Vec2 *
    P_minus) {
    double lam = 1.0 / sqrt(a * a + b * b);
    P_plus->x = lam * a * a;
    P_plus->y = lam * b * b;
```

C code

```
P_minus->x = -P_plus->x;
P_minus->y = -P_plus->y;
}

// ----- Compute tangent line points (parametric) -----
void tangent_line_points(Vec2 n, Vec2 N, double span, int npts,
    double *x_out, double *y_out) {
    // J @ n = (n.y, -n.x)
    double tx = n.y;
    double ty = -n.x;
    double step = (2 * span) / (npts - 1);
    for (int i = 0; i < npts; i++) {
        double s = -span + i * step;
        x_out[i] = N.x + tx * s;
        y_out[i] = N.y + ty * s;
    }
}
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from ctypes import CDLL, Structure, c_double, c_int, POINTER

# Load shared library (after compiling)
lib = CDLL("./formula.so")

# Define Vec2 struct in Python
class Vec2(Structure):
    _fields_ = [("x", c_double), ("y", c_double)]

# Function prototypes
lib.compute_P_points.argtypes = [c_double, c_double, POINTER(Vec2),
                                  POINTER(Vec2)]
lib.tangent_normal_at.argtypes = [c_double, c_double, Vec2,
                                   POINTER(Vec2)]
lib.foot_from_origin_to_line.argtypes = [Vec2, c_double, POINTER(Vec2)]
lib.area_triangle.argtypes = [Vec2, Vec2, Vec2]
```

```
lib.area_triangle.restype = c_double
lib.tangent_line_points.argtypes = [Vec2, Vec2, c_double, c_int,
                                     POINTER(c_double), POINTER(
                                         c_double)]

# Parameters
a, b = 3.0, 2.0
npts = 200
span = 6.0

# Create needed structs
P_plus, P_minus = Vec2(), Vec2()
lib.compute_P_points(a, b, P_plus, P_minus)

# Tangent normals
n1, n2 = Vec2(), Vec2()
lib.tangent_normal_at(a, b, P_plus, n1)
lib.tangent_normal_at(a, b, P_minus, n2)
```

Python Code

```
# Feet from origin
N1, N2 = Vec2(), Vec2()
lib.foot_from_origin_to_line(n1, 1.0, N1)
lib.foot_from_origin_to_line(n2, 1.0, N2)

# Area check (optional)
O = Vec2(0.0, 0.0)
area1 = lib.area_triangle(P_plus, O, N1)
area2 = lib.area_triangle(P_minus, O, N2)

# Tangent line points
x1 = (c_double * npts)()
y1 = (c_double * npts)()
x2 = (c_double * npts)()
y2 = (c_double * npts)()
lib.tangent_line_points(n1, N1, span, npts, x1, y1)
lib.tangent_line_points(n2, N2, span, npts, x2, y2)
```

Python Code

```
# Convert arrays to NumPy
tx1, ty1 = np.array(x1), np.array(y1)
tx2, ty2 = np.array(x2), np.array(y2)

# Ellipse points
theta = np.linspace(0, 2*np.pi, 600)
x_ellipse = a * np.cos(theta)
y_ellipse = b * np.sin(theta)

# ----- Plot -----
fig, ax = plt.subplots(figsize=(8,8))
ax.plot(x_ellipse, y_ellipse, label='Ellipse', linewidth=2)
ax.scatter([0.x], [0.y], color='k', s=40, label='0 (origin)')
ax.scatter([P_plus.x, P_minus.x], [P_plus.y, P_minus.y], color='
    tab:blue', s=80, label='P (optimal)')
ax.scatter([N1.x, N2.x], [N1.y, N2.y], marker='x', s=90, color='
    tab:orange', label='N (foot)')
ax.plot(tx1, ty1, '--', linewidth=1.2, label='Tangent at P')
```

```
ax.plot(tx2, ty2, '--', linewidth=1.2)

# Triangles
ax.fill([0.x, P_plus.x, N1.x], [0.y, P_plus.y, N1.y], alpha=0.25)
ax.fill([0.x, P_minus.x, N2.x], [0.y, P_minus.y, N2.y], alpha
        =0.25)

# Labels
ax.text(0.x+0.1, 0.y+0.1, f"0({0.x:.2f},{0.y:.2f})", fontsize=11)
ax.text(P_plus.x+0.1, P_plus.y+0.1, f"P({P_plus.x:.2f},{P_plus.y
    :.2f})", fontsize=11)
ax.text(P_minus.x+0.1, P_minus.y+0.1, f"P({P_minus.x:.2f},{
    P_minus.y:.2f})", fontsize=11)
ax.text(N1.x+0.1, N1.y+0.1, f"N({N1.x:.2f},{N1.y:.2f})", fontsize
    =11)
ax.text(N2.x+0.1, N2.y+0.1, f"N({N2.x:.2f},{N2.y:.2f})", fontsize
    =11)
```

```
x.set_aspect('equal', 'box')
ax.set_xlim(-2*a, 2*a)
ax.set_ylim(-2*b, 2*b)
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Ellipse and triangles PON (from C via .so)')
ax.grid(True)
ax.legend(loc='upper left')

plt.show()
```

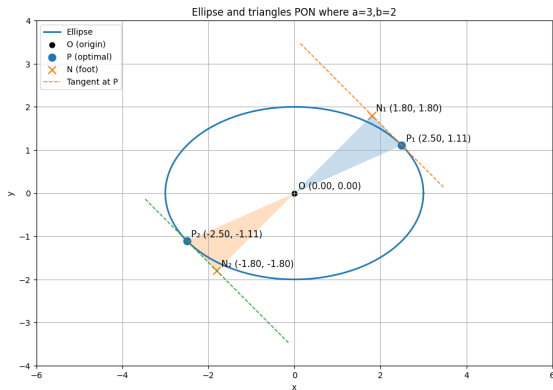


Figure: Ellipse and triangles PON where $a=3, b=2$